

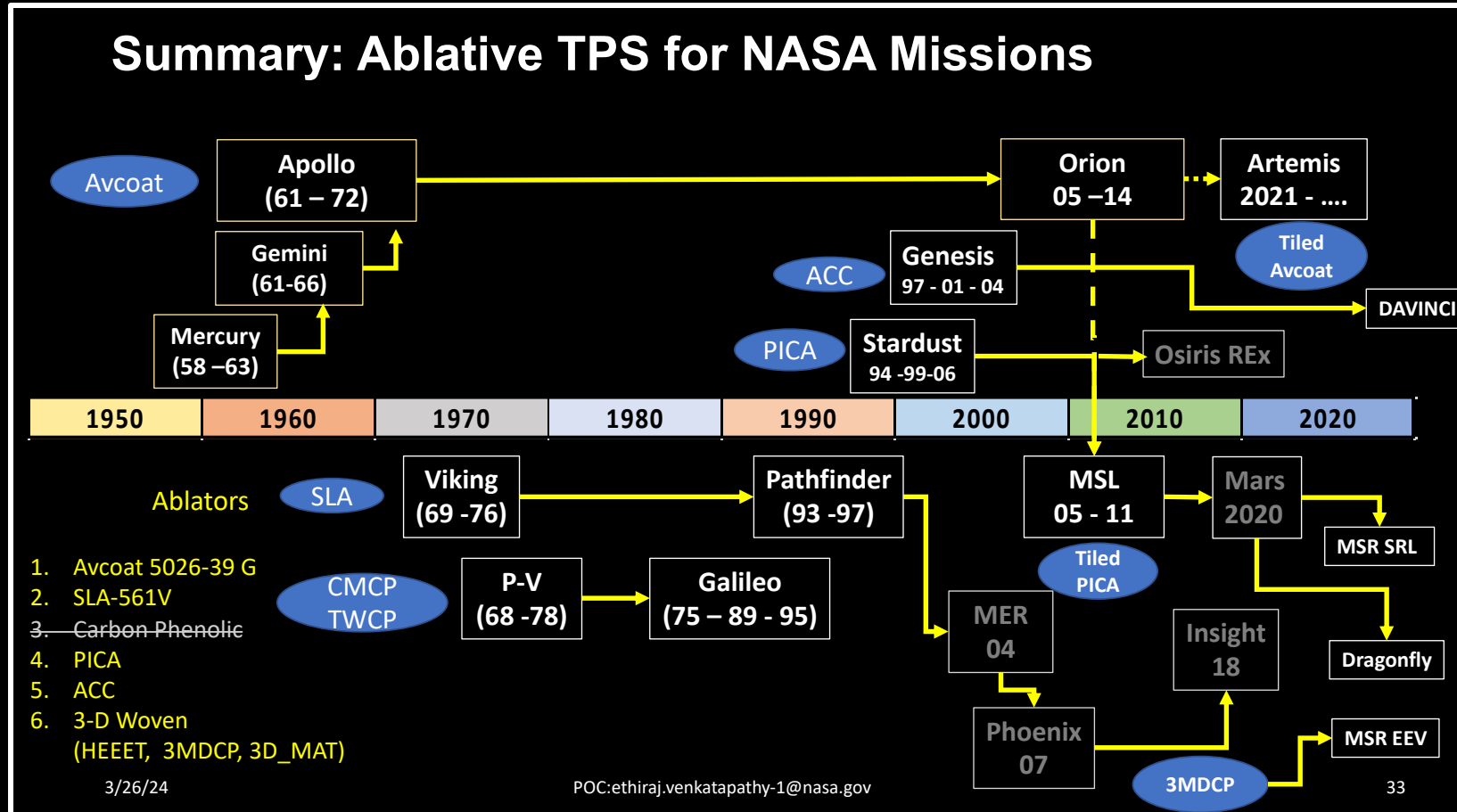
History of Ablative TPS used in NASA Missions

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What is This Talk about?

It is about thermal protection materials and systems (TPS) developed and used in NASA missions.



Supersonic, Hypersonic and Reaching Orbit (1940s – 1950s)

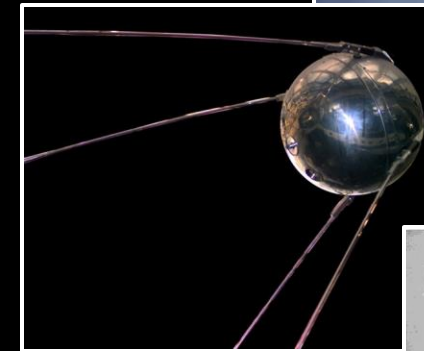
- WWII - V-2 Ballistic Missile (1942)
 - Rockets were detonating prematurely due to re-entry heating
 - Plywood heatshields solved the heating problem
 - Discipline of TPS began
- First hypersonic flight was the two-stage Bumper rocket (1949)
 - Consisted of a second stage set on top of a V-2 first stage
 - Reached Mach 6.7, but burned-up /destroyed during entry
- Sputnik 1, first orbital flight of a spacecraft (1957)
 - Orbited for three weeks; batteries died; after two months. during reentry (by design) burned/destroyed
- X-15, first successful hypersonic flight at Mach 6 (1961)
 - Inconel and Titanium metallic structure protected from hypersonic heating
 - In addition, Spray-on silicone-based ablator for acreage and asbestos/silicone moldable TPS for leading edges
- ICBMs with ablative TPS successfully tested (1959)



V2-Rocket



X-15

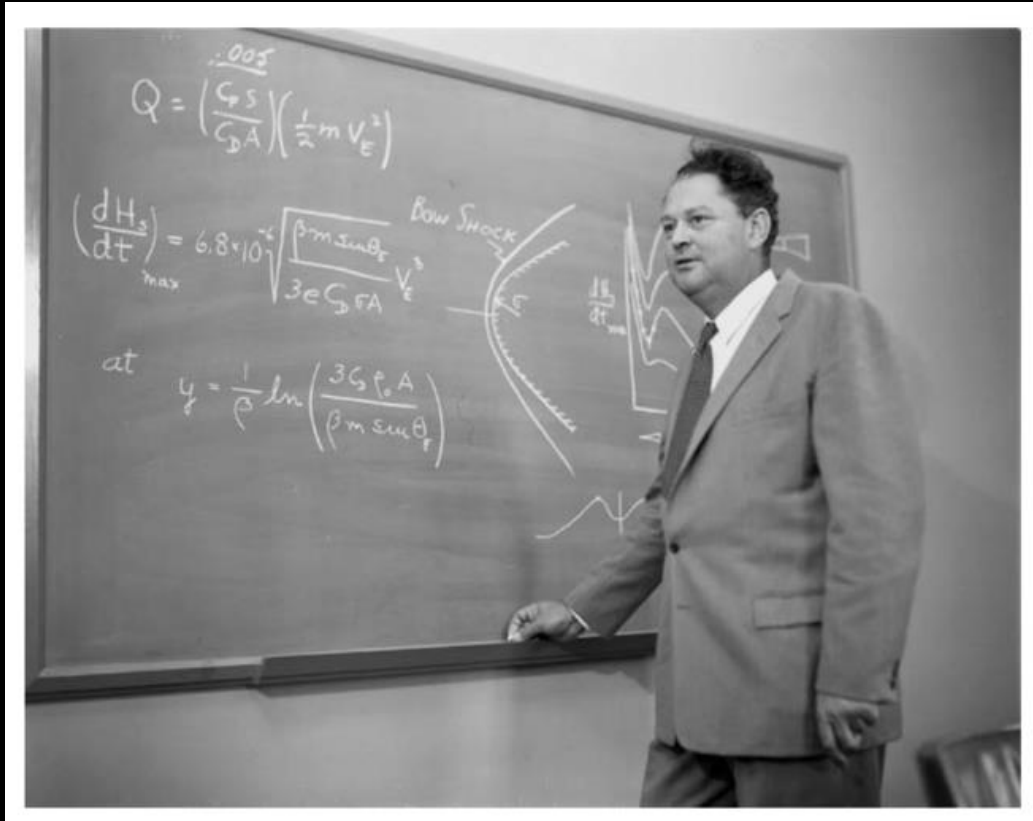


Sputnik 1



Reentry Vehicles 3

Blunt Body: A Revolutionary Concept called Blunt-body and Ablative TPS - Solution to Hypervelocity Entry (1950s)



Harvey Allen (1952), Blunt-body Concept for surviving Hypersonic Reentry

“Atmospheric drag is the most efficient way to slow down. But need protection from entry heating”

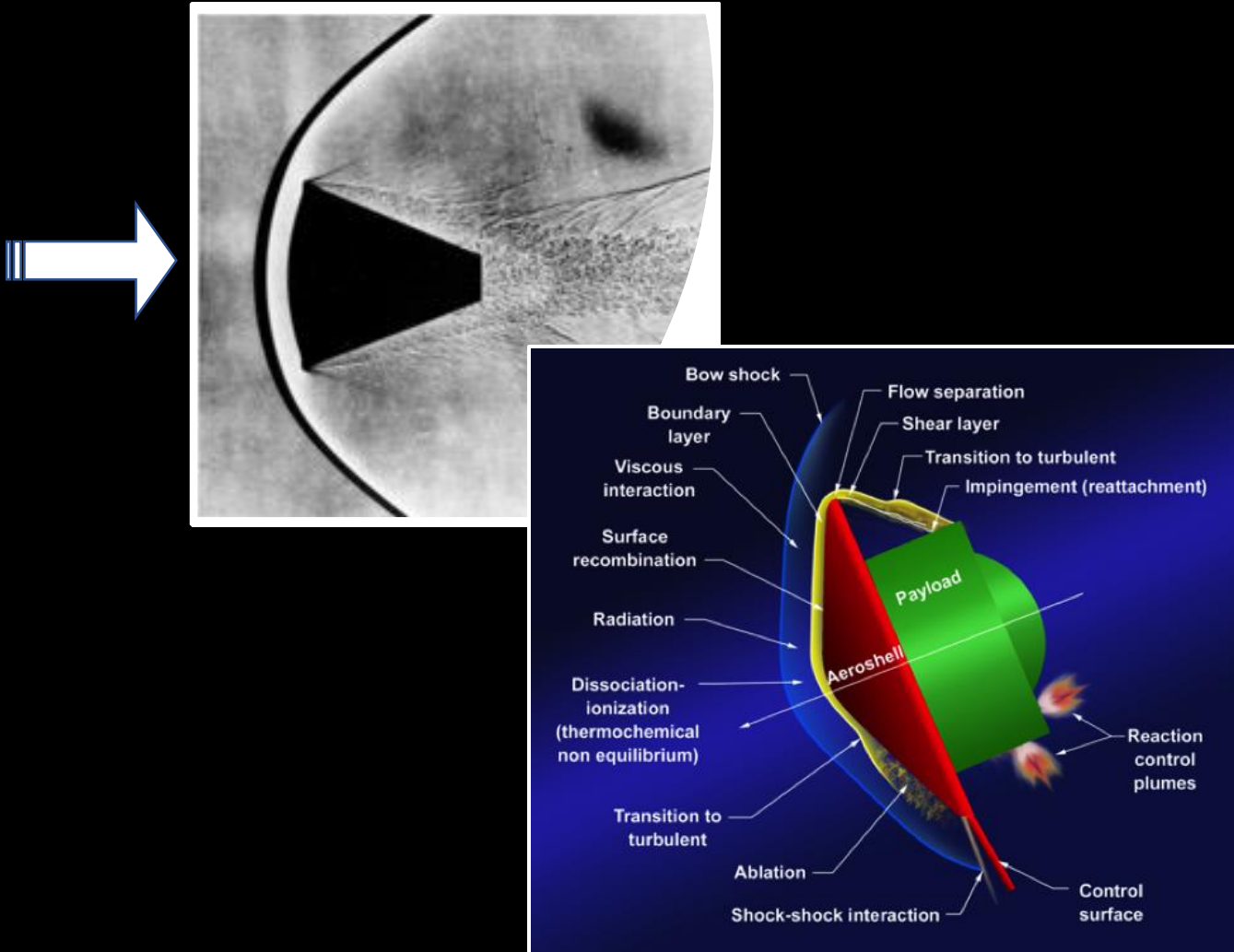
“Half the heat generated by friction was going into the missiles.”

“I reasoned we had to deflect the heat into the air and let it dissipate. Therefore, streamlined shapes were the worst possible; they had to be blunt.”

-- Harvey Allen

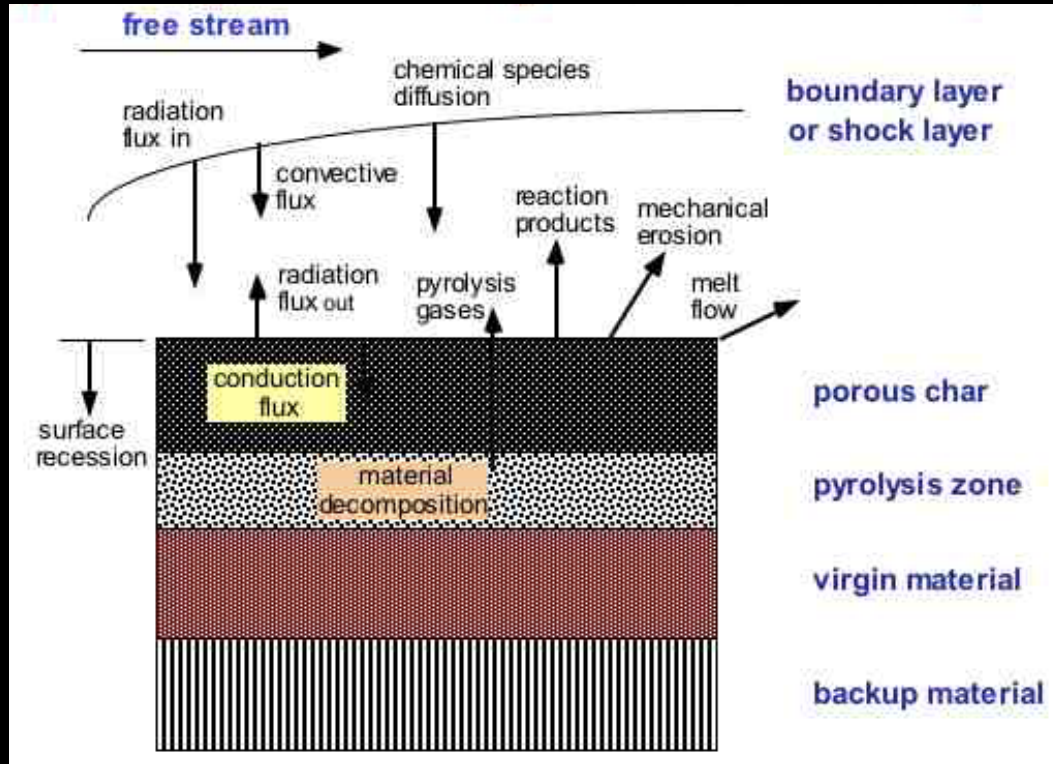
Fail-safe thermal protection system design requires comprehensive understanding of the hypervelocity, reacting flow (aerothermodynamics), and selection, design, testing and verification of TPS as part of the integrated system.

Hypersonic Flow



- Molecules bunch up against the body and get compressed
- Shock wave forms
- Compression causes heating
- Air-molecules break up into atoms releasing more energy
- Temperatures (10,000– 20,000) K
- The spacecraft slows down enough
- Heating happens only for a short duration during entry

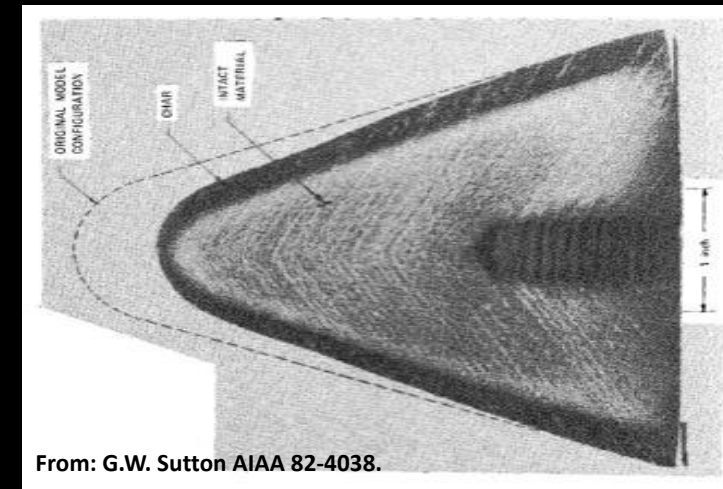
Dealing with Heat: Ablative Thermal Protection System



Ablative material keeps the extreme heat away from the body for a short time

Choosing Material for Rejecting Heat

- Material that forms good char and radiates heat away
- Material that is insulative (don't let the heat in)
- Material that does not spall or flake away
- Resin that can "boil off" to keep the heat away



From: G.W. Sutton AIAA 82-4038.

Post arc-jet test article
Wrapped Silica-Phenolic (1958) (1 inch base)

Apollo Missions (1960 – 1970)

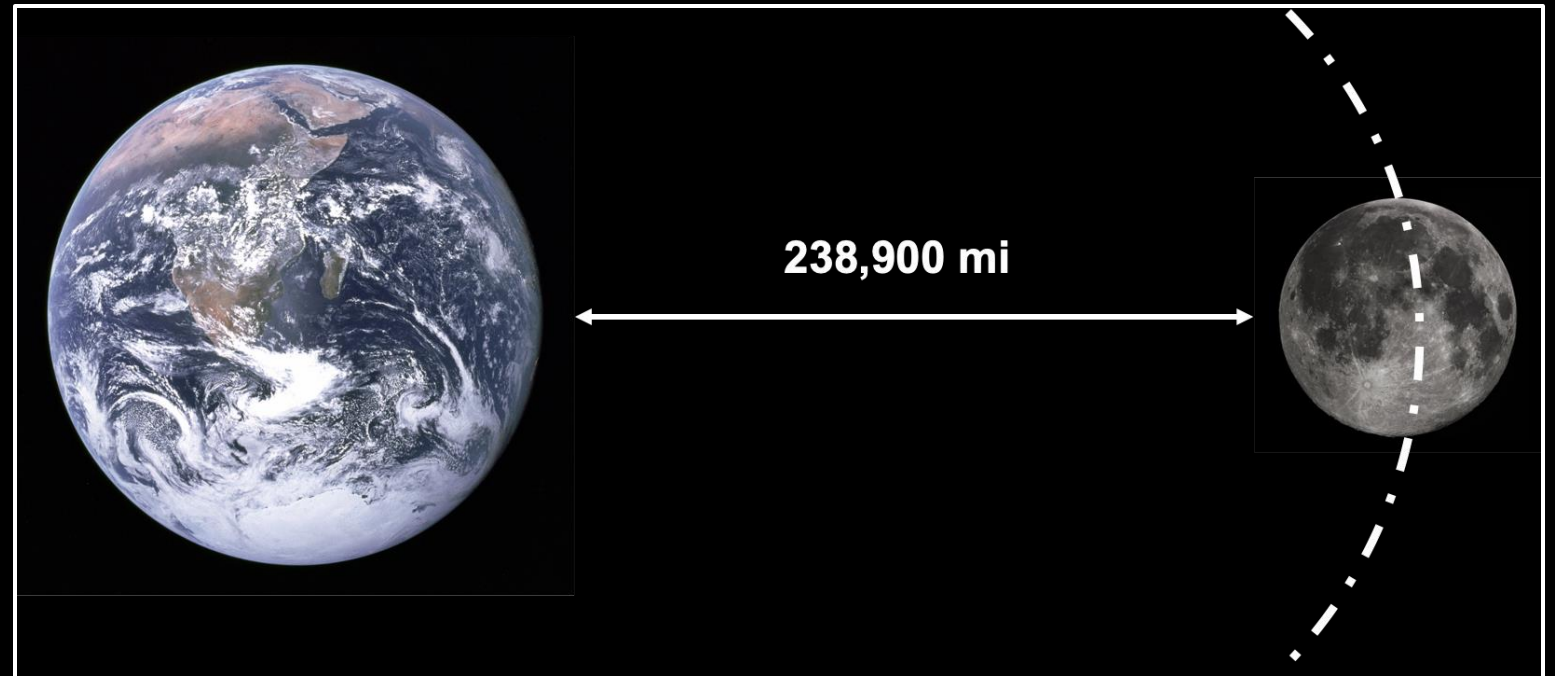
To Land Astronauts on the Moon and Bring them Back Safely

363 ft Tall and 33 ft Dia.



Image: NASA

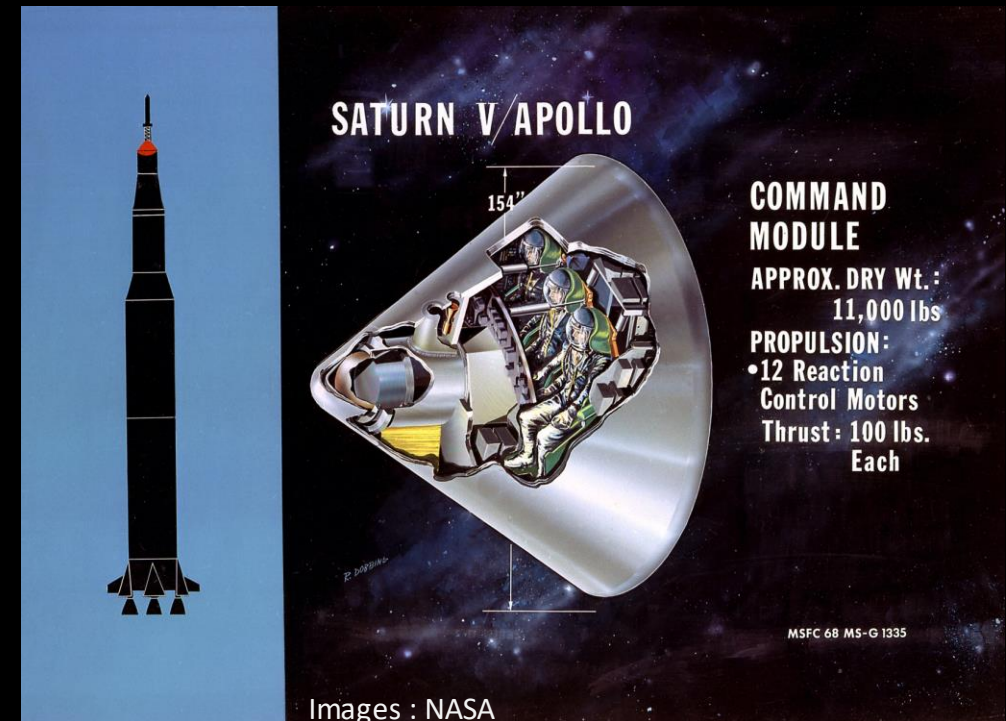
Payload	107, 100 lbs
Lift off Mass	6,540,000 lbs



Apollo speed ~25,000 mph and slowed down in less than 7min.

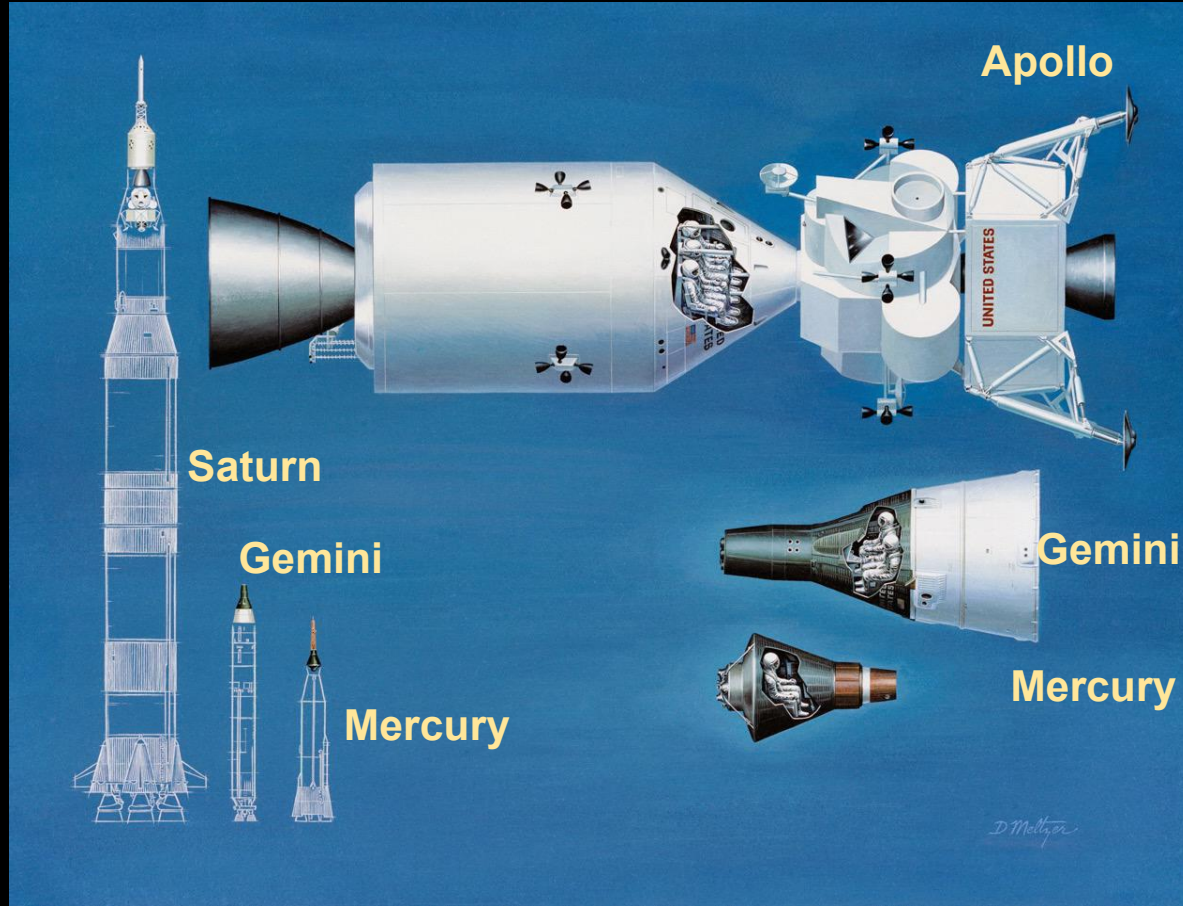
How to Slow Down and Land?

- At the Moon, we use a retro-rocket to slow down from 3,600 mph to touchdown
- Coming back from Moon, the Command Module will be traveling at 25,000 mph.
- Could we use a Retro-Rocket?
 - The Retro-Rocket must be carried all the way to the Moon and back.
- The size of the new rocket will be so enormous that it will be Impossible to build and launch in the first place.



**We figured out how to use the atmosphere to slow down.
But then we had to deal with heating that comes with it.**

Landing on the Moon: Mercury, Gemini and Apollo Accelerated Development



Overlapping Developments

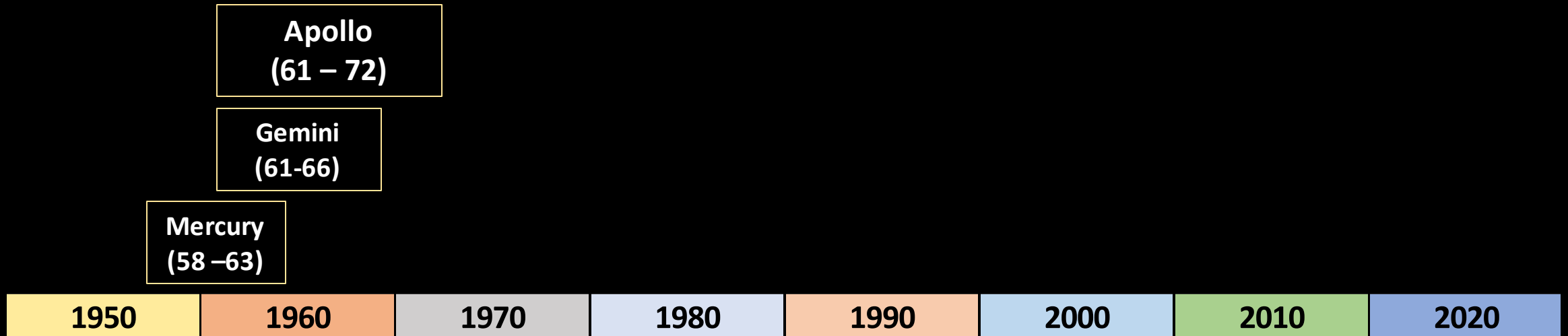
- Mercury (1959 – 1963)
- Gemini (1961 – 1966)
- Apollo (1961 - 1972)

Covered all aspects – from Launch to Return

Learning, on the fly, through testing



Mercury, Gemini and Apollo (50's and 60'):



Note: Kennedy announced his goal of putting a man on the moon and bring them back safely by the end of the decade on May 25, 1961.

Mercury (1959 – 1963) 20 Un-crewed and 6 crewed flights

1.9 m heat shield – Peak heating ~ 50 W/cm²



- January, 1959 - thermal protection options
 - Beryllium heat sink or an ablator.
- Fiberglass-phenolic shingles for the heat-shield.
 - Beryllium and Rene metallic for the conical sections
- **Feb 20, 1962: John H. Glenn Jr. became the first American to orbit the Earth on Mercury.**

System Integration:

- Early test of heatshield separation resulted in recontact – Decision was made to keep the heat-shield intact
- Retro rockets attached to the heat-shield.
- Water impact landing

Manufacturing

- Heatshield center plug did not bond well and kept falling off on most manned flights.
- Center plug missing on Friendship 7. Number of cracks in HS exterior, did not compromise mission safety.



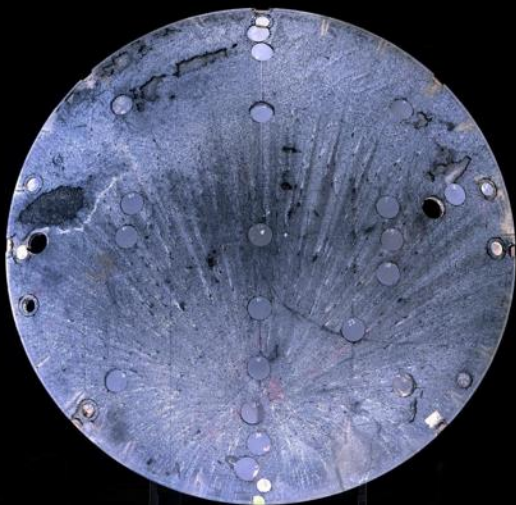
Heat shield sample from Scott Carpenter's "Aurora 7" spacecraft (May 24, 1962)

Gemini (1961 – 1966) 2 Un-crewed and 10 crewed flights

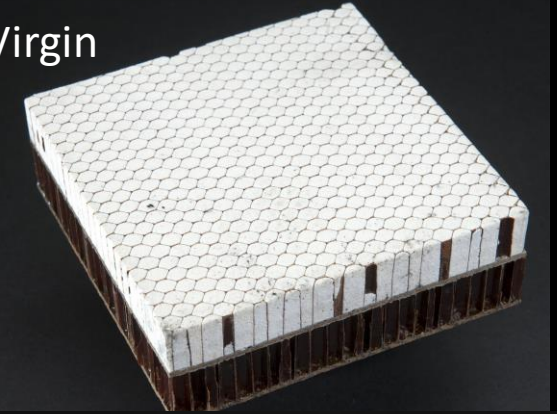
3.05m heat shield – Peak heating ~ 60 W/cm²



- Ablative material is fiberglass honeycomb filled with silicone elastomer.
 - Corning DC 235 Silicone elastomer
- Honeycomb bonded to the structure allowed attachment verification
- The ablative substance is a paste-like silicone elastomer material which hardens after being poured into a honeycomb form.
- Room temperature cure
- Virgin white surface turns black as a result of charring.
- Glass forms at the surface during heating and melt flow



Virgin



Heat shield fragment from the Gemini VIII, which carried astronauts Neil Armstrong and David Scott into Earth orbit on March 16, 1966

Apollo (1961 - 1972) 16 Un-crewed and 11 Crewed flights

3.91 m heat shield – Peak heating ~ 600 W/cm²



- Avco Corp. developed the ablative material, and the heat shield manufacturing process in 3 years.
- Avcoat 5026-39G. Epoxy-novalac resin reinforced with quartz fibers and phenolic micro-balloons in a phenolic honeycomb
- 360,000 cells. Avco invented a way to fill each of the cells by hand and developed repair procedures.
- Qualified for In-space thermal environment of (-260°F - + 250°F)
- Impact of Micro-meteor were assessed
- Forebody penetration (Compression pad) and other singularities



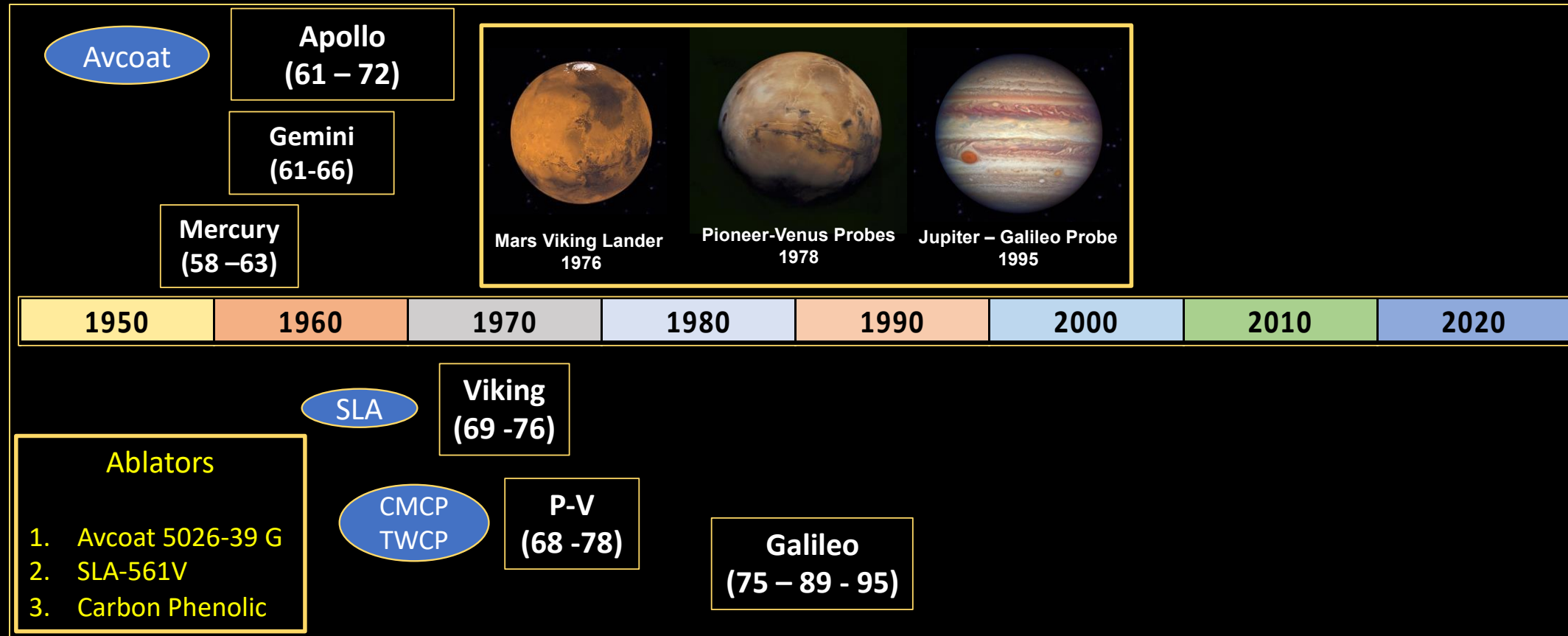
Apollo (1961 - 1972) – AVCO Corp movie now available on You Tube “1000 Seconds to Home: Apollo Heat Shield”



<https://www.youtube.com/watch?v=64UM3CUqSfg>

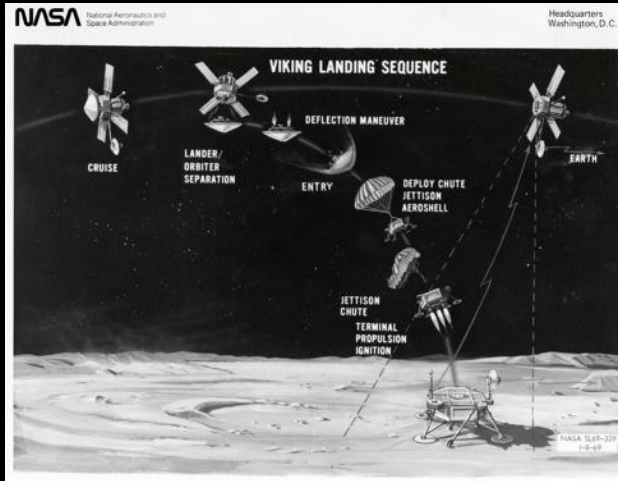
History of Ablators for NASA Missions (1970 – 1980):

In December 1968, NASA created the Space Shuttle Task Group to determine the optimal design for a reusable spacecraft, and issued study contracts to General Dynamics, Lockheed, McDonnell Douglas, and North American Rockwell. January 1972, President Nixon approved the Shuttle. The science community, leveraging Apollo success, has been eager to launch missions to Mars, Venus and Giant Planets before Shuttle.



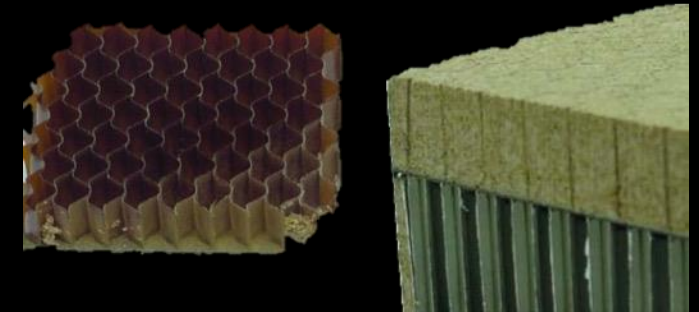
Viking Landers (1976) 2 Landers – Delivered from Orbit

3.5m heat shield – Peak heating ~ 26 W/cm²



InSight Heat Shield Assembly – Photo: Lockheed Martin

- Super-Lightweight Ablator - SLA-561V
 - Apollo heatshield too heavy for Mars entry
 - Good insulator and lighter (< 50% the density of Apollo)
- Developed (late 60's) by Martin Marietta*
 - Made of ground cork, silica and carbon fibers, silica and phenolic micro-balloons and silicone.
 - Similar to Apollo ablator, uses honeycomb to bond to the structure and hold the ablative material.
- Manufacturing
 - Hand packing. Could not use “gunning.”
- Testing showed robust performance
 - In space environment, vibro-acoustics, thermo-structural, thermal
- Planetary protection requirement



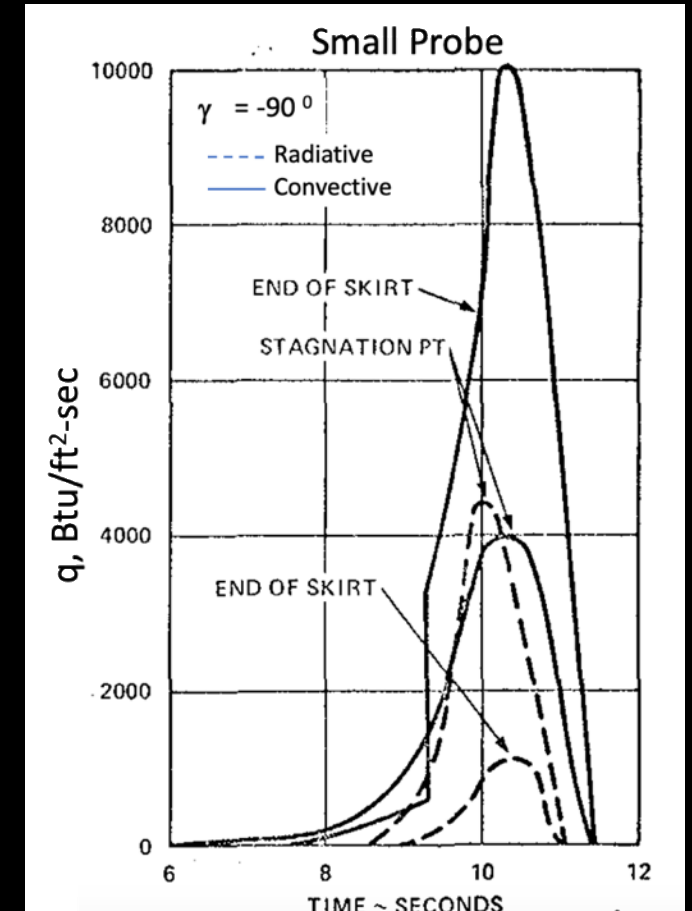
Pioneer-Venus (1978) 3 Small (0.76 m) and 1 Large (1.52 m) Probes

Peak design heat flux 10 kW/cm² at 10 atm. pressure

Pioneer-Venus:



- 4 probes (3 small and one large)
- Shock layer radiation dominated (much more severe than Apollo)
- Combined convective and radiative - an order of magnitude higher than Apollo
- NASA did not have materials to handle severe entry conditions for the Venus
 - Reflective material (Teflon) considered but rejected
 - Phenolic-Nylon vs Carbon-Phenolic
 - Carbon-Phenolic tested in air and CO₂
- GE (Valley Forge) developed high density carbon phenolic, two-piece, heat shield (1975)
 - Two different manufacturing, nose and frustum with Chop-Molded and Tape-Wrapped carbon phenolic
 - Tape-wrapped derived from DoD

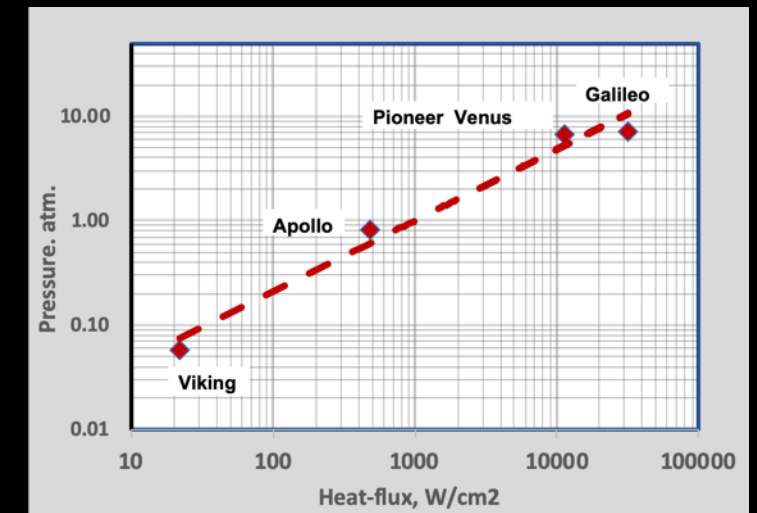
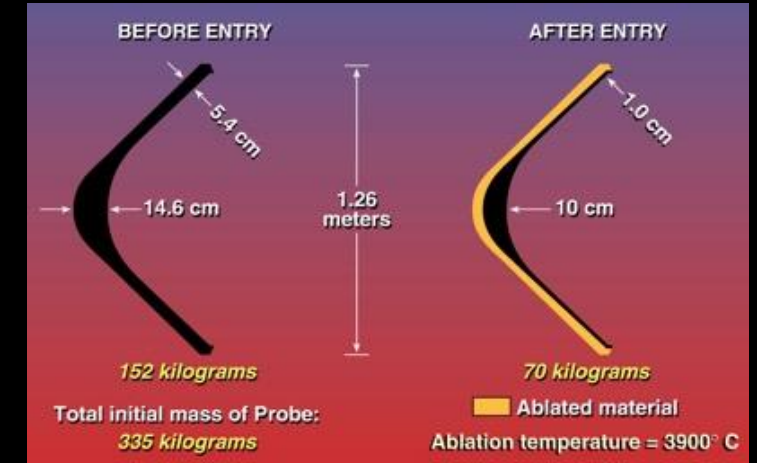


Galileo (1995) - Single Probe Entry at 47.4 km/sec

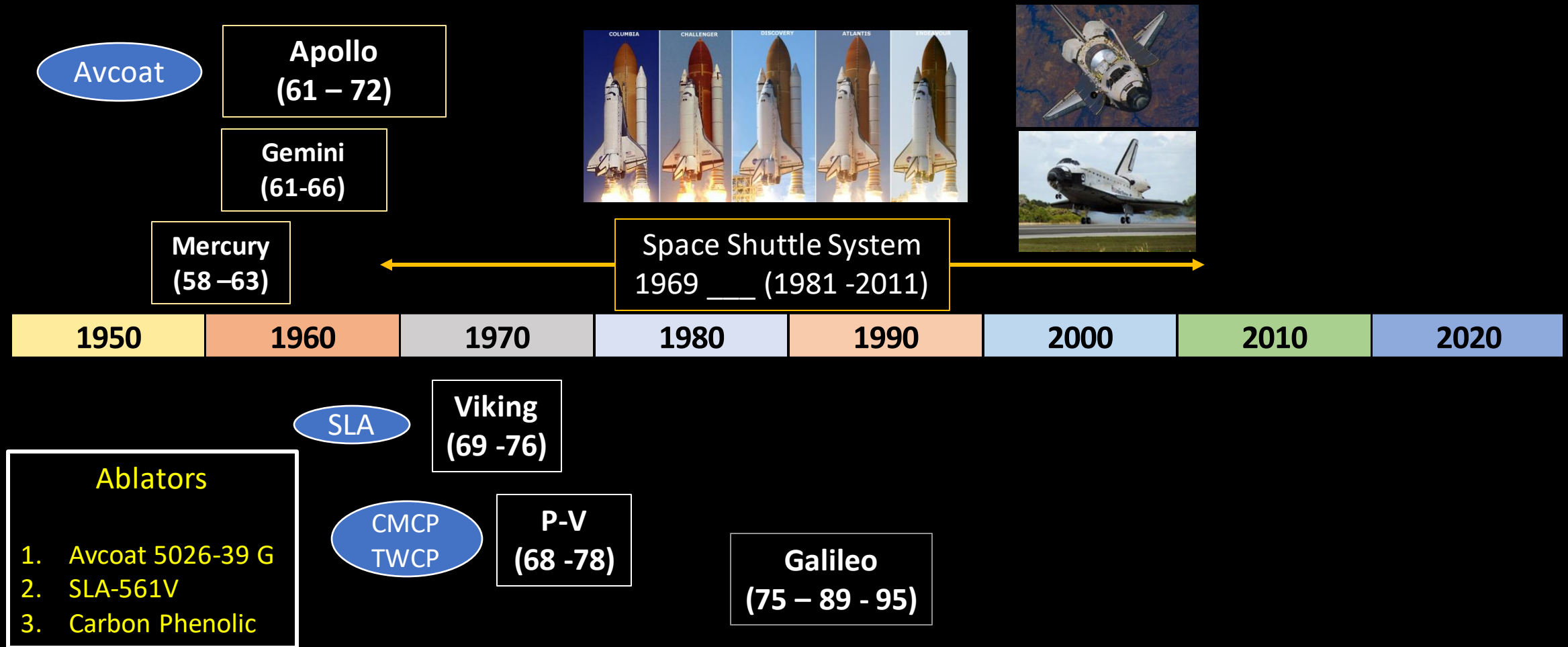
Peak conditions (30,000 W/cm² and 7.3 atm)



- Predicted heating at ~ 49 km/sec entry – extreme
- Galileo peak heat-flux exceeded the heating at the nose tip of a ballistic missile and the radiative heating from a thermonuclear explosion combined.
- GE Carbon Phenolic (2 piece) chop molded and tape wrapped heat shield manufacturing from P-V.
- Heat shield mass was 50% of entry mass.
- Recession sensors in the heat shield provided data
- In 15 sec., 50% of the heat shield ablated
- Margined design proved to be non-conservative on the shoulder – near failure
- Based on flight measurement, margined TPS will be 70% of the entry mass

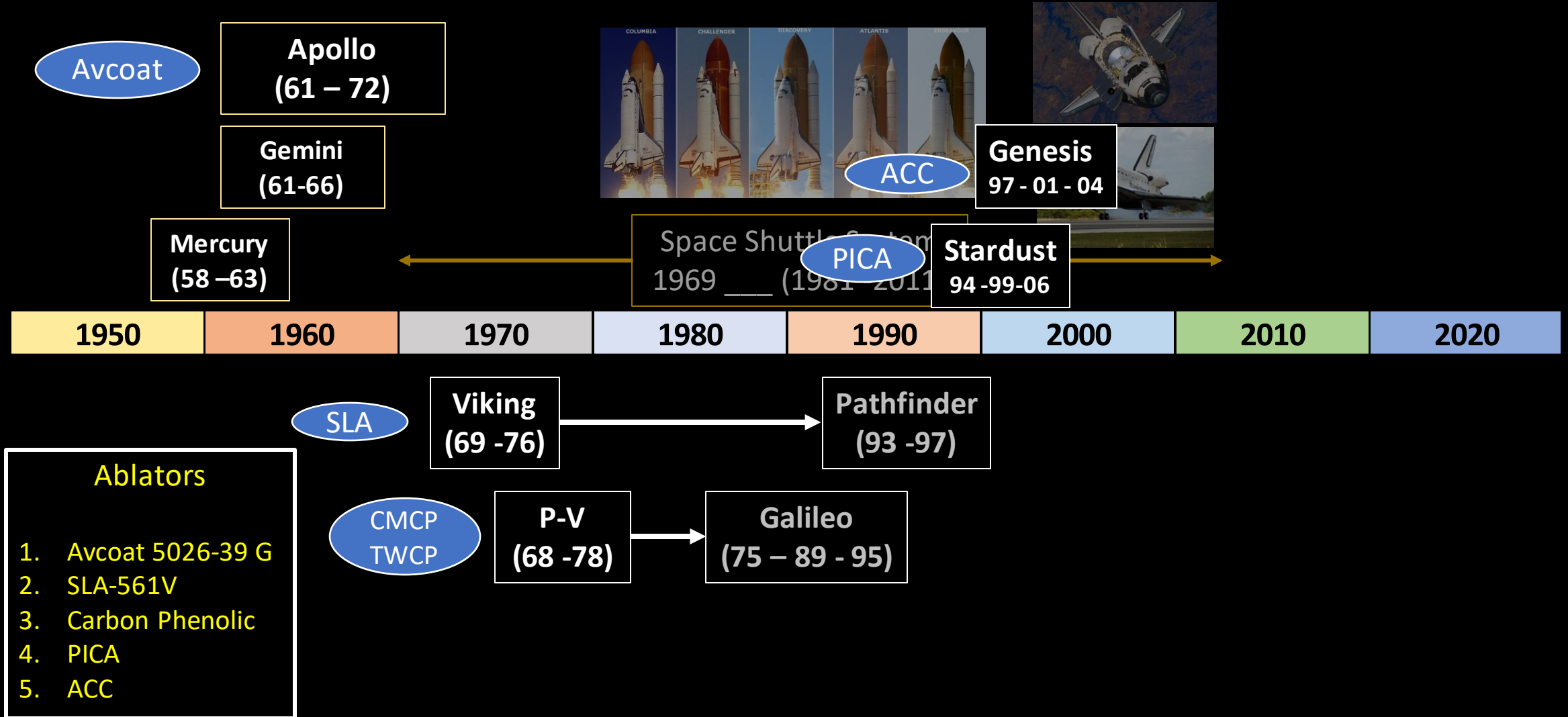


Reusable TPS for Space Shuttle Orbiter (1980 - 2010):



In December 1968, NASA created the Space Shuttle Task Group to determine the optimal design for a reusable spacecraft, and issued study contracts to General Dynamics, Lockheed, McDonnell Douglas, and North American Rockwell. January 1972, President Nixon approved the Shuttle.

History of Ablators for NASA Missions (1990s):



Pathfinder (2.65m) , Stardust (0.83) and Genesis (1.5) (90's) Smaller “Faster, Better, Cheaper” Missions

- Pathfinder mission with small Sojourner rover,
 - Landed successfully on Mars on July 4, 1997
 - SLA-561V (recovered from Viking era) heat shield
- Stardust sample return capsule at 12.5 km/sec
 - Launched in 1999 returned to Earth Jan. 15, 2006
 - Phenolic Impregnated Carbon Ablator (PICA)
 - Single piece, seamless heat shield
 - Very capable ($> 1000 \text{ W/cm}^2$ and $> 0.5 \text{ atm}$ pressure)
 - Low density (0.25 g/cc) ~ SLA561-V density but capable of much higher entry
- Genesis
 - Launched in 2001, returned in 2004 - Parachute failed
 - PICA manufacturing could not be scaled up in time
 - Advanced Carbon-Carbon + Low density carbon insulation
 - Seamless but with attachment points on the heat shield

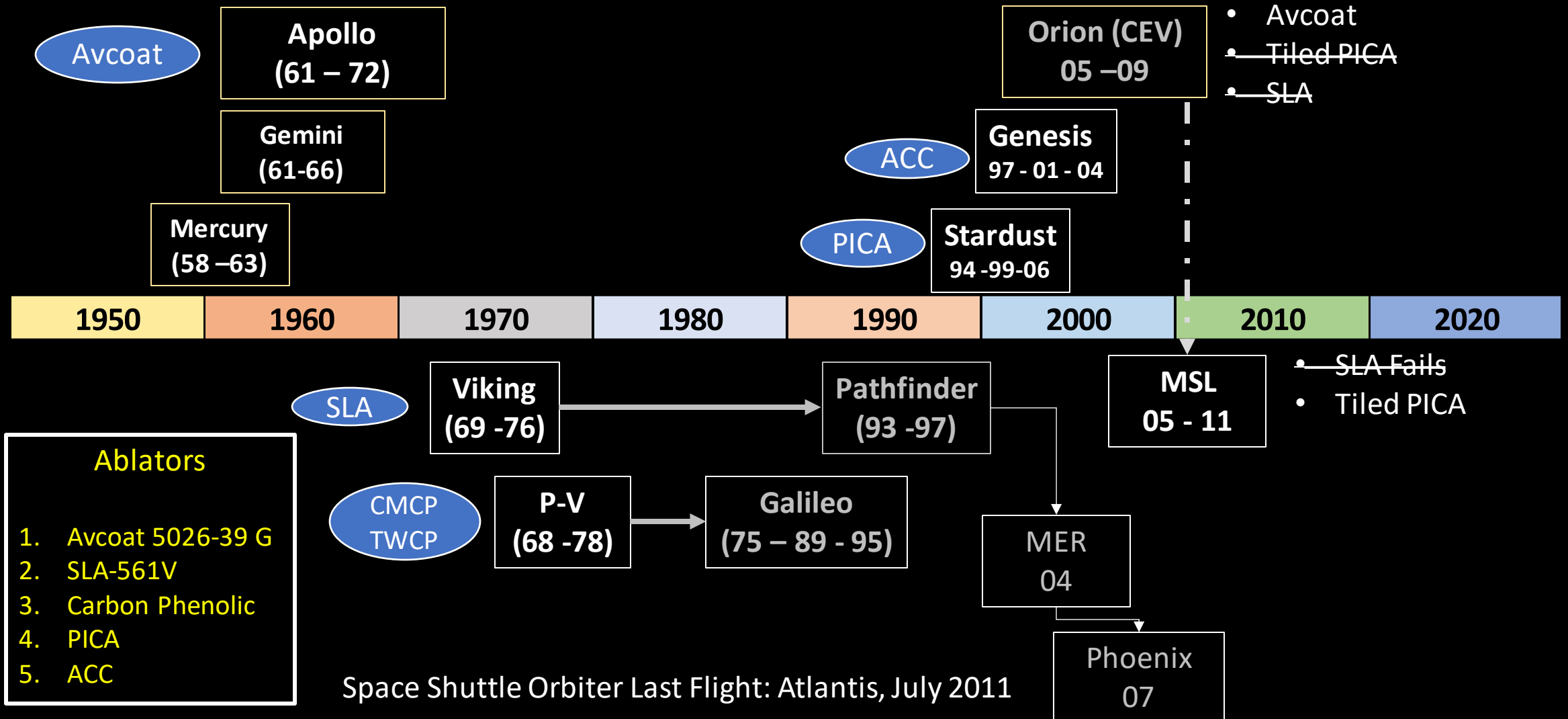


<https://www.jpl.nasa.gov/who-we-are/media-information/press-kits>



Diameter: Stardust 0.81m ; Genesis: 1.5m

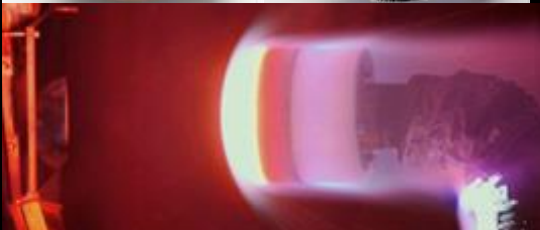
History of Ablators for NASA Missions (2000 - 2010):



To the Moon – Once Again

Orion to Artemis (2005 - 2009 - 2014 – 2022) EM1....

SLA-561 V Wedge Test In the Arc Jet



- The largest ablative Heat Shield ever made
- Avcoat™ 5026-39 HC/G (Honeycomb-Gunned)
 - Epoxy novolac resin in a fiberglass honeycomb matrix
 - > 300,000 Avcoat™ cells
- 5 m (16.5 ft) diameter
- 1,800 kg (4,000 lb) including carrier structure
 - ~450 kg (1,000 lb) of Avcoat



Post flight-tested (EFT-1) Shield

Mars Science Laboratory (MSL) Switched from SLA-561V to Tiled PICA in 2008



- Launched December 2011
- Successfully Delivered Curiosity on August 6, 2012
- Mars 2020, built to print PICA heatshield
- Mars SRL planned to use PICA with lyocell - a rayon replacement
- Dragonfly has baselined and will use PICA-D (domestic Lyocell)

Artemis Missions: Artemis-1 (2022)

Switched from H/C Avcoat to Tiled Avcoat

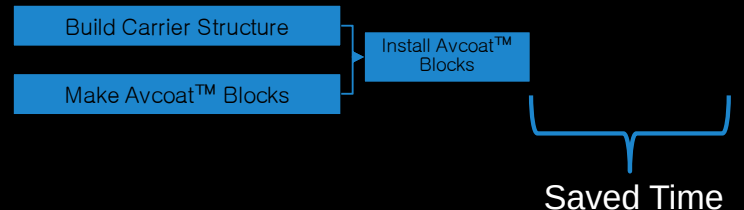
- NASA decided to switch from H/C and fly a block Avcoat with gap-fillers
- Technical: Difficulties closing the HC/G architecture with positive stress margins
 - EFT-1 challenges –how to improve manufacturing and the basic material strength
 - EM mission loads are greater than EFT-1
- Programmatic: Difficulties fitting the HC/G architecture into the Program's schedule "box"
 - HC/G architecture manufacturing is a serial process
 - Block architecture manufacturing is a semi-parallel process



HC/G Timeline:

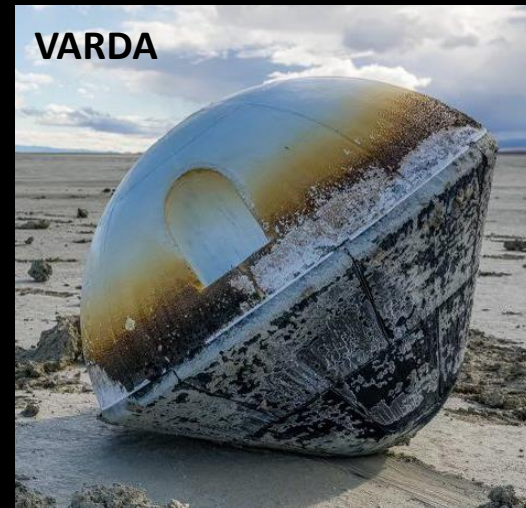


Molded Block Timeline:



Successful Commercial Missions

- **Space X Dragon 1 and Dragon 2 (Crew and Cargo)**
 - PICA-X (PICA) and PICA-3 (Conformal PICA) 5-m diameter
 - Launched 2010, Crew 1 May 2020
- **VARDA**
 - Conformal PICA, 1-m diameter
 - Launched 2023/Re-entered Feb 2024
- **Emerging Commercial Space has a continued need for TPS**
 - Heritage Options as well as new developments, low cost
 - Ablative as well as reusable



VARDA

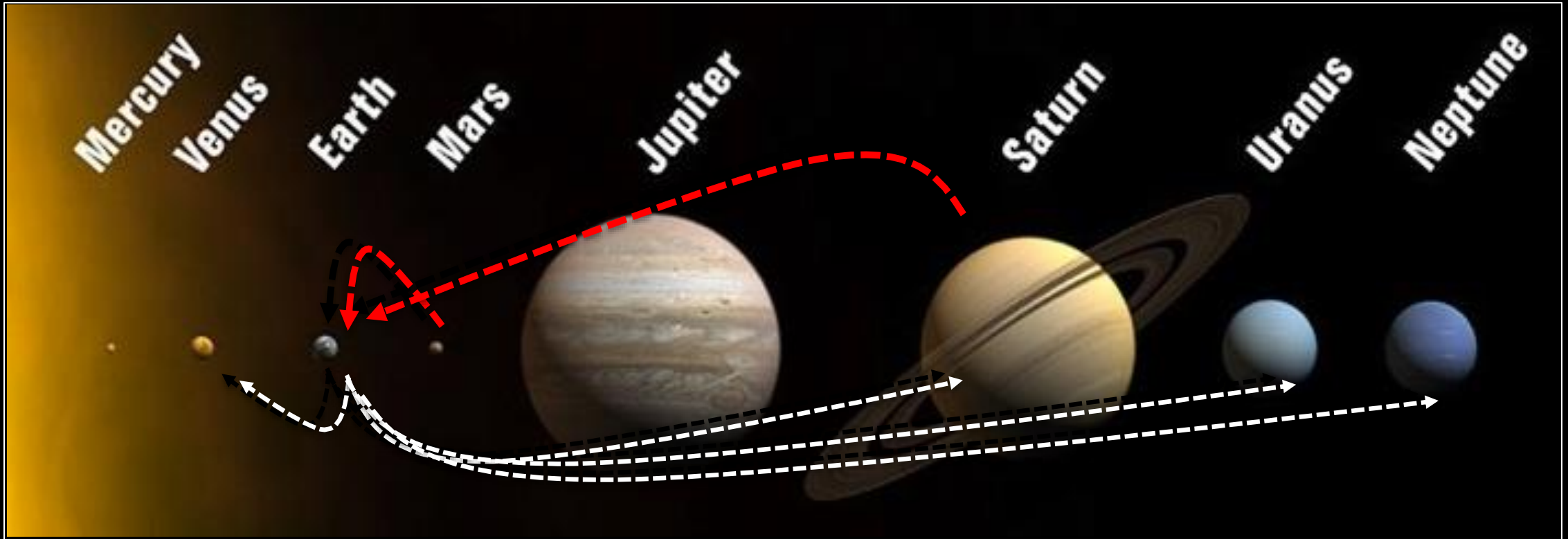


SpaceX Dragon

<https://www.scientificamerican.com/article/nasa-astronauts-splashdown-safely-after-historic-spacex-mission/>

<https://www.space.com/var-da-in-space-manufacturing-capsule-reentry-photos>

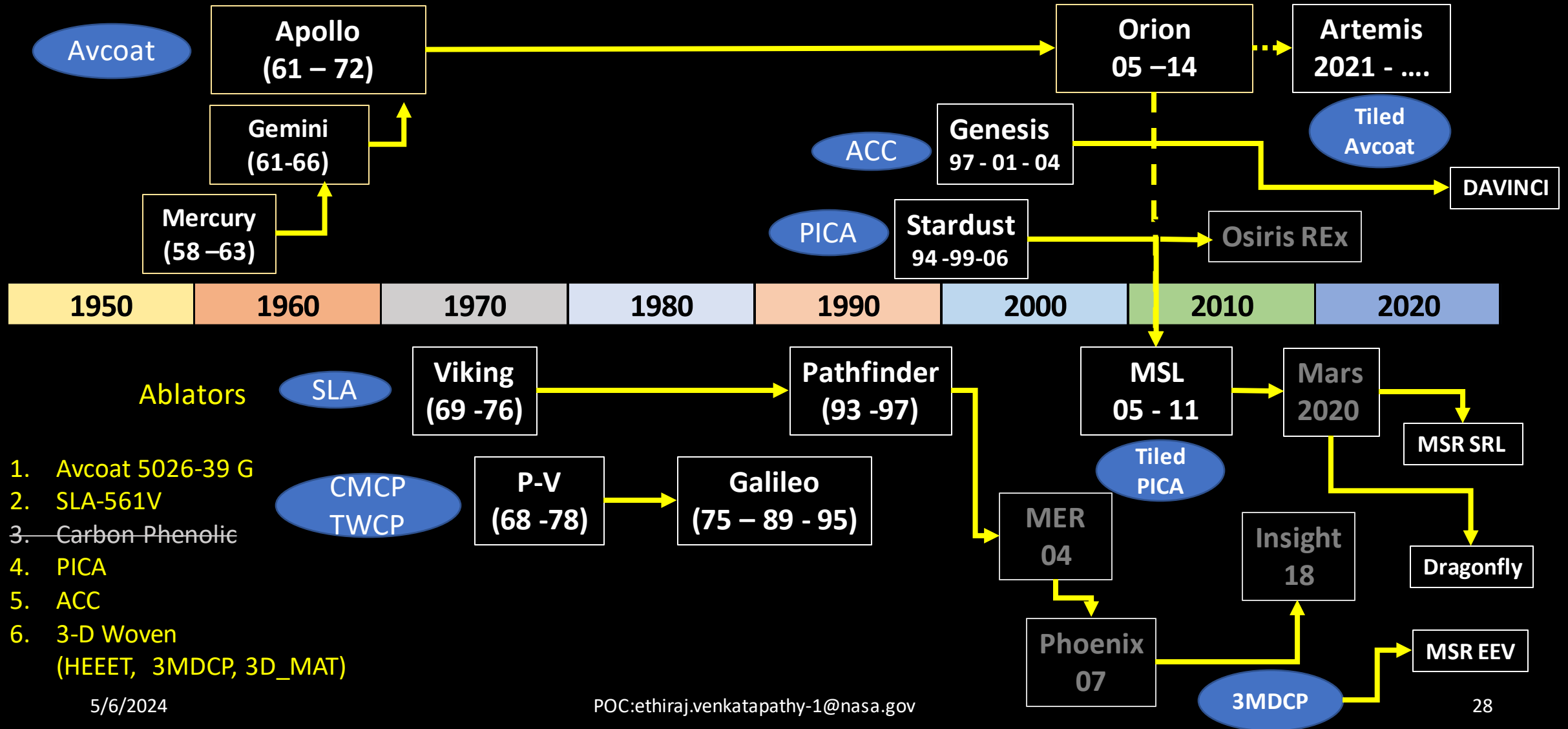
Decadal Survey (2013 and 2023) Recommended Science Missions (Missions 2030 and Beyond)



Moon, Mars and Beyond

- In-Situ Exploration of Venus, Saturn, Ice Giants, Sample Return from Mars and Water Worlds
- Cislunar and Human Mars Mission

Summary: Ablative TPS for NASA Missions



Concluding Remarks

- NASA, in the past 60+ years, has successfully executed many planetary missions that demanded benign to extreme entry and TPS was essential and enabling.
- Ablative thermal protection system is agnostic to human or robotic science missions.
- TPS sits in the middle and requirements including risk drives the design and testing for a specific mission. Often flight experience from previous missions are leveraged but there are challenges every mission poses.
- **The number of flight proven TPS is very few**

Acknowledgements

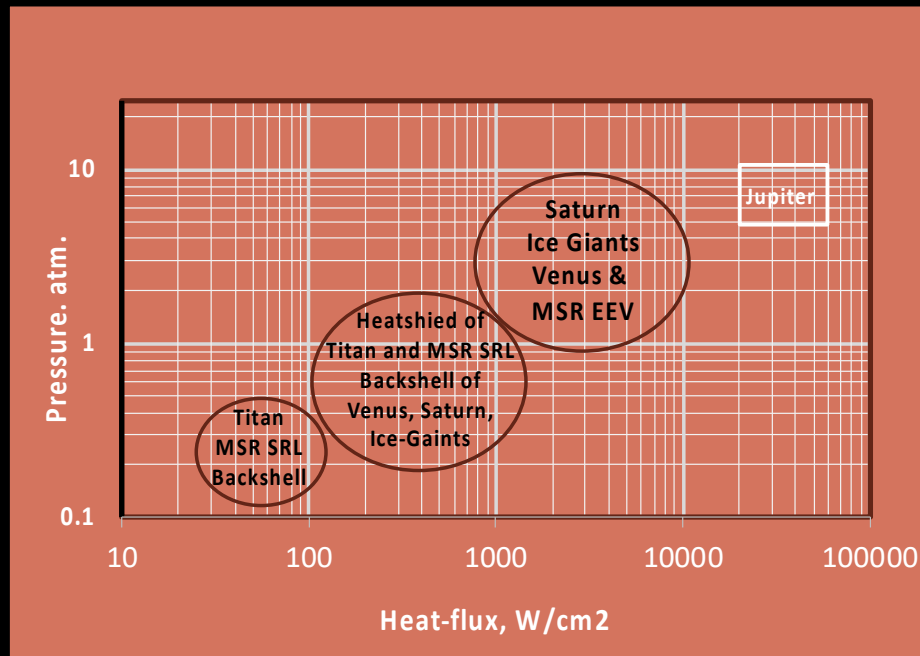
We sincerely wish to thank our colleagues at NASA, Industry, Academia and elsewhere, too many to name here; for teaching, mentoring and learning with me over the years.

Our colleagues at NASA Ames are very special and our gratitude to all of them.

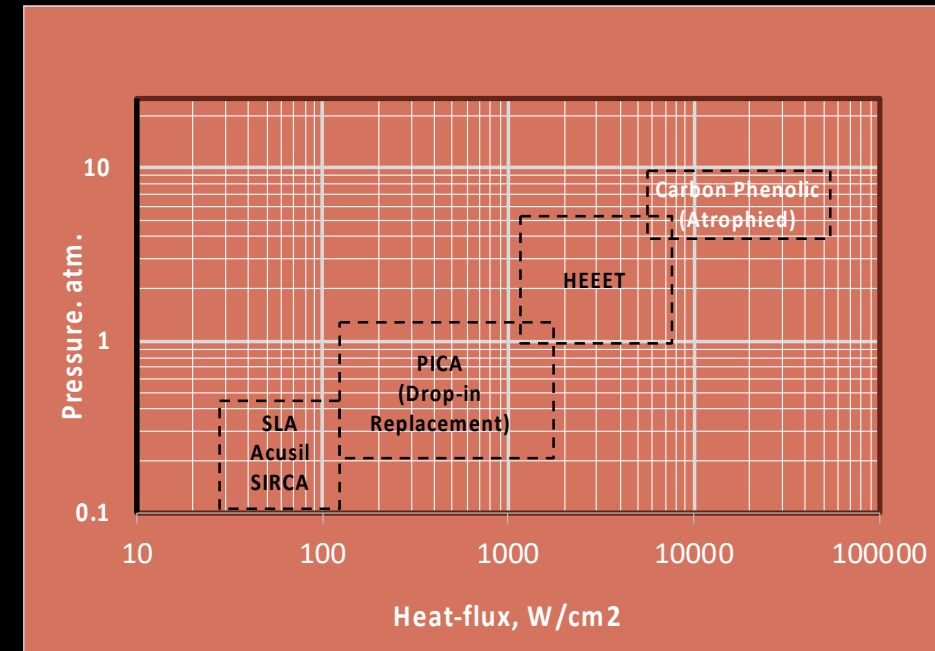
NASA Developed Ablative TPS Capability vs Future Mission Needs:

There are few mature TPS capable of meeting the needs of future missions (development of new or alternates is a decade long process)

Entry Environment



CAPABILITY



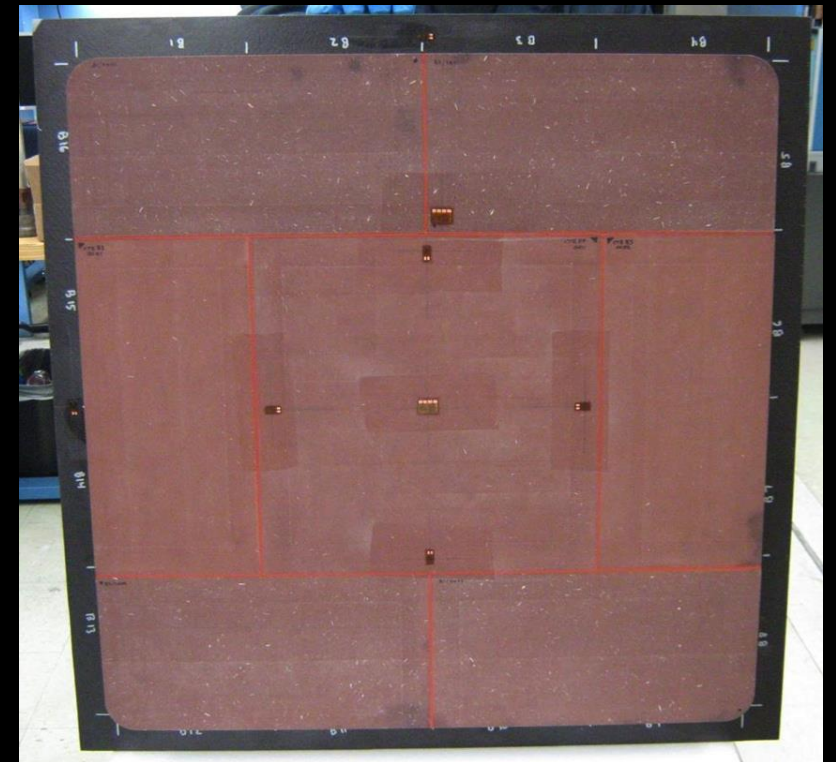
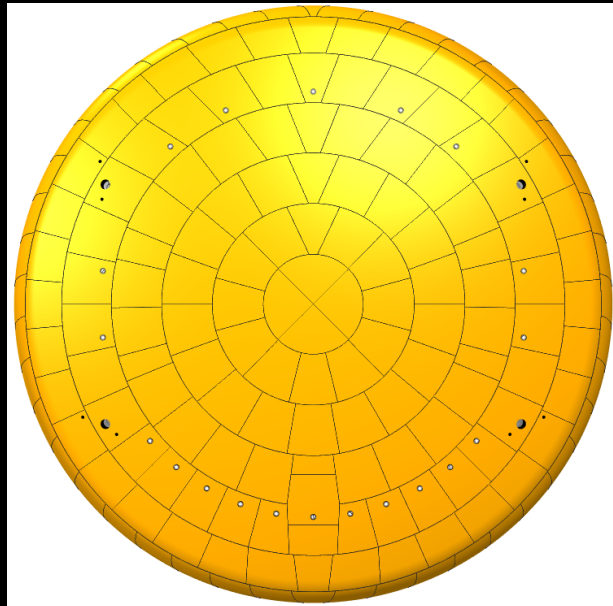
- HEEET and PICA, two NASA developed TPS, are needed and need to be sustained for majority of future SMD-PSD missions

The Block Architecture

Heritage?: Space X Dragon



- ~ 300 Avcoat™ blocks bonded to the carrier structure
 - Same ablator, molded into blocks instead of gunned into a honeycomb
- RTV seams



Advantages and Disadvantages of Blocks Relative to the HC/G System

- **Advantages**

- “True” acceptance testing and as-built ablator inspection
- Parallel manufacturing with the carrier structure
- Cheaper fabrication cost (no manual gunning, no honeycomb operations)
- Faster test article production throughput
- Higher ablator strength
- Less density variations

- **Disadvantages**

- New system element: block interfaces
- Less capable attachment system
- Less tolerant to first failure modes
- Differential recession at block interfaces – can cause a mass increase if seam fences or gaps cause heating augmentation